Sheet: VeraModule

VERA FPGA

File: vera-fpga.sch

Sheet: BusDecoder

BUS DECODER

File: busdecoder.sch

Sheet: Vera FPGA flash

VERA SPI FLASH
SD CARD INTERFACE

File: vera-fpga-flash.sch

Sheet: CartridgeInterface

CARTRIDGE INTERFACE

File: cartridgeInterface.sch

Sheet: PowerSupply

POWER SUPPLY

File: powersupply.sch

Sheet: VGA Analog

VGA ANALOG

File: vga-analog.sch

FIDUCIAL TOP

FID3 Fiducial

FID4 Fiducial

FIDUCIAL BOTTOM

FID2 Fiducial

FID1 Fiducial

LOGO1
VERA X16 LOGO

LOGO2
ATARI READY

LOGO3
KiCAD DESIGN

LOGO4
ATARI DUO BUS

LOGO5
ESP32_BOBOARD

LBL1

H1
MountingHole

H2
MountingHole

H3
MountingHole

H4
MountingHole

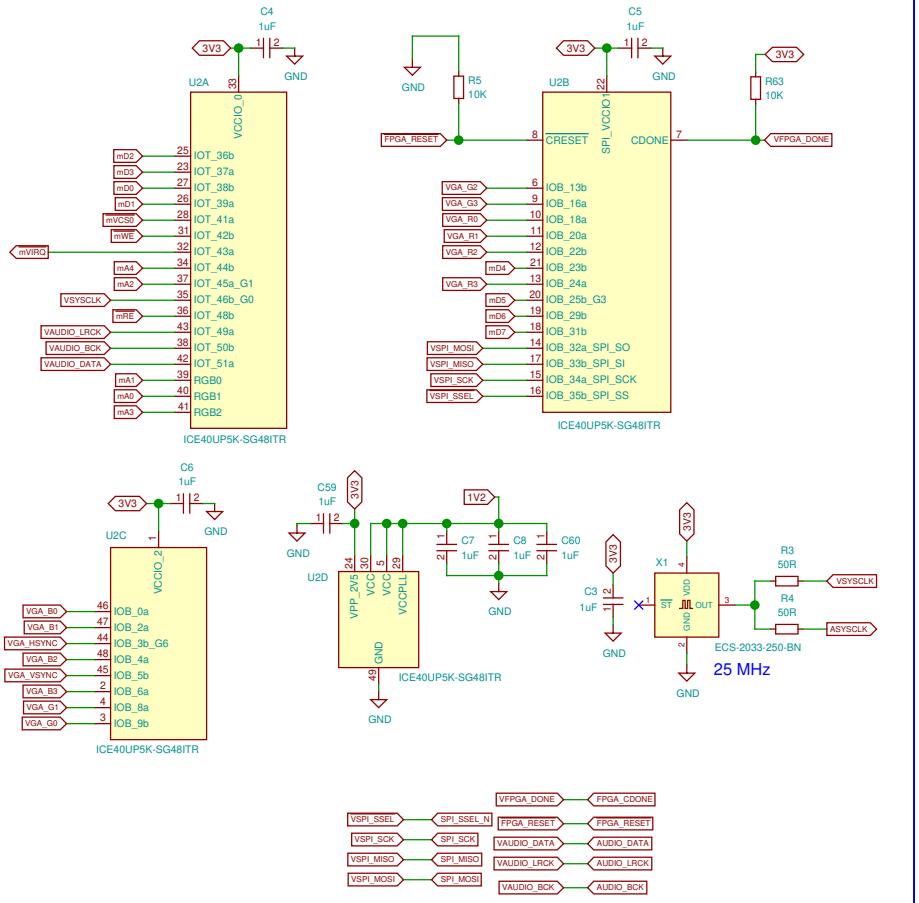
Gianluca Renzi
RetroBit Lab

Sheet: /
File: VERA-MODULE-RBL.sch

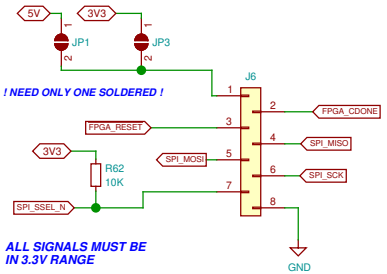
Title: VERA FPGA Audio & Video Board

Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261029-20:38Z		Id: 1/7

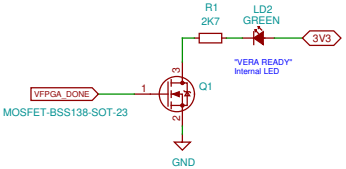
FPGA VERA LOGIC VIDEO & AUDIO CARD



iceprog programmer USB FTDI / SPI

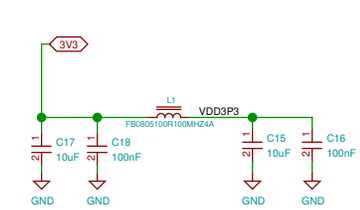


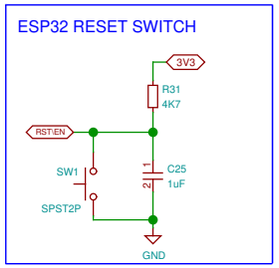
VERA FPGA PROGRAMMED OK

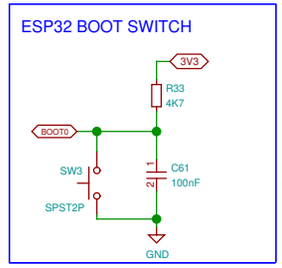


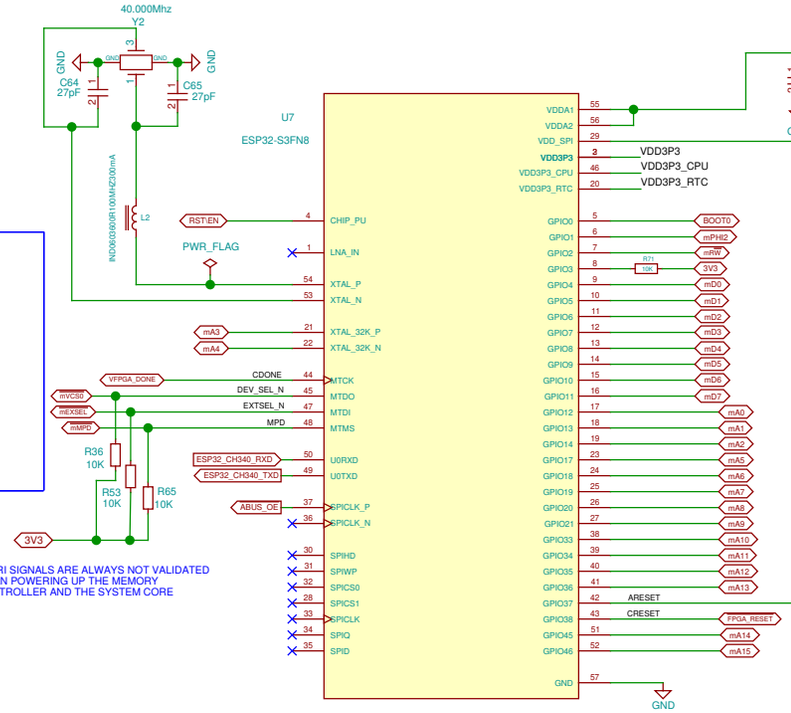
Gianluca Renzi	
RetroBit Lab	
Sheet: /VeraModule/	
File: vera-fpga.sch	
Title: VeraModule	
Size: A4	Date: 2026-10-26
Rev: 1.0	
Plotted by eeshow 3a954ac+ 20261029-20:38Z	Id: 2/7

PBI DEVICE ID: software selectable only



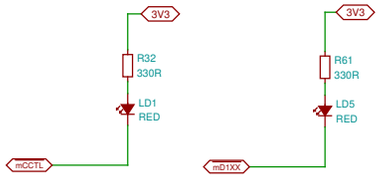






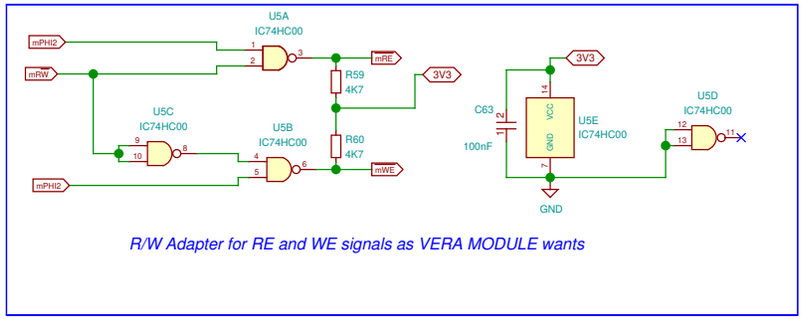
HARDWARE DRIVEN LEDS

Accessing $D5XX area space

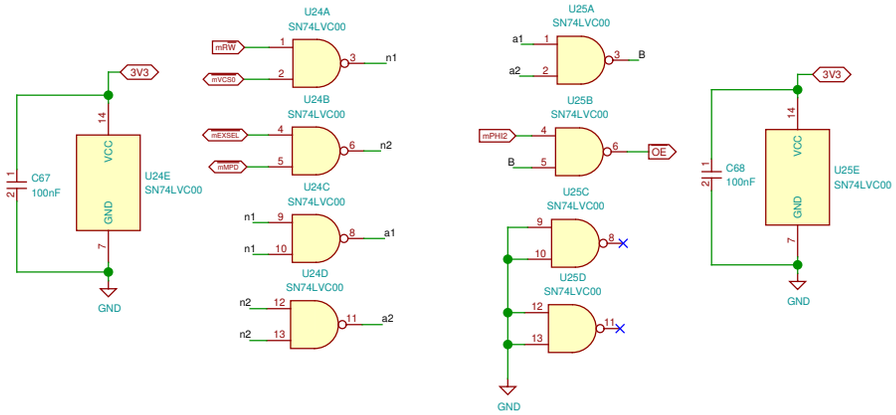


Accessing $D1XX area space

PBI Bus Interface Decoder:
I/O Space, Math Pack Disable, ROM ADDRESS ($D800-$DFFF),
RAM PBI SPACE ($D600-$D7FF)



R/W Adapter for RE and WE signals as VERA MODULE wants



When VERA Chip Select (mVCS0) latch is active & A15..A0 $D800-$DFFF -> MPD active (External 2K ROM)
When VERA Chip Select (mVCS0) latch is active & A15..A0 $D600-$D7FF -> EXSEL active (External 512 bytes RAM)
When access @ $D1xx and VERA Chip Select (mVCS0) latch is active -> EXSEL active

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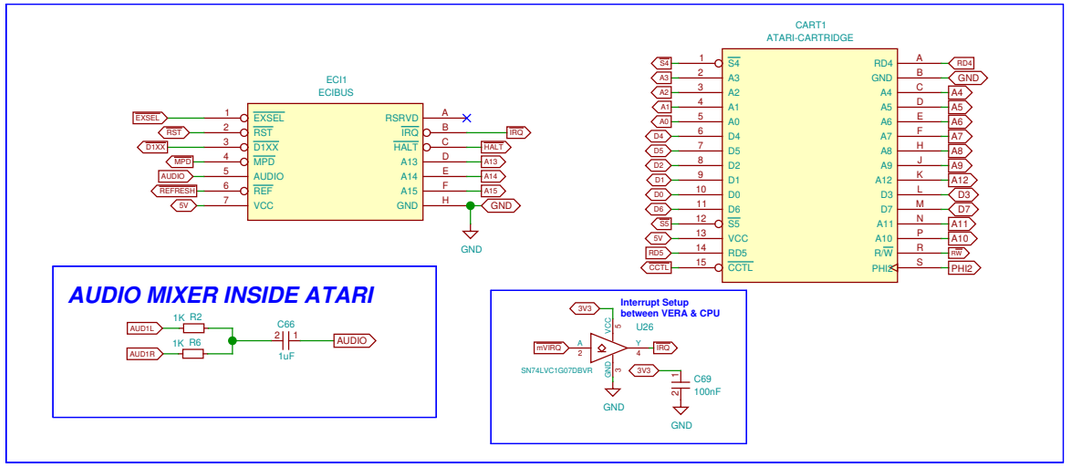
Sheet: /BusDecoder/
File: busdecoder.sch

Title: BusDecoder

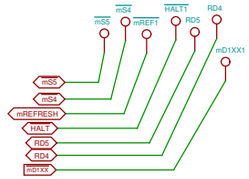
Size: A3	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261029-20:38Z		Id: 3/7

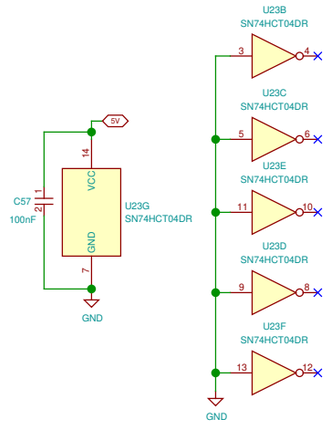
ATARI 130XE ECI & CARTRIDGE INTERFACE

All 8-Bit signals must be shifted from 5V to 3.3V and vice versa

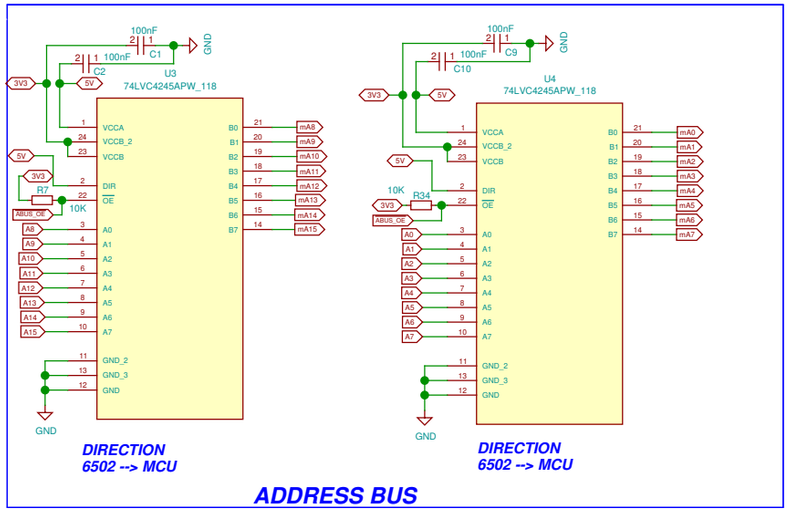


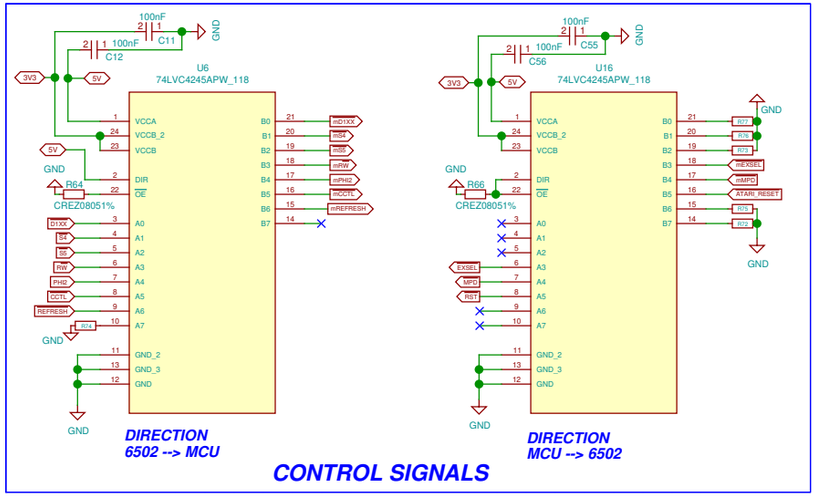
UNCONNECTED IN THIS PROJECT

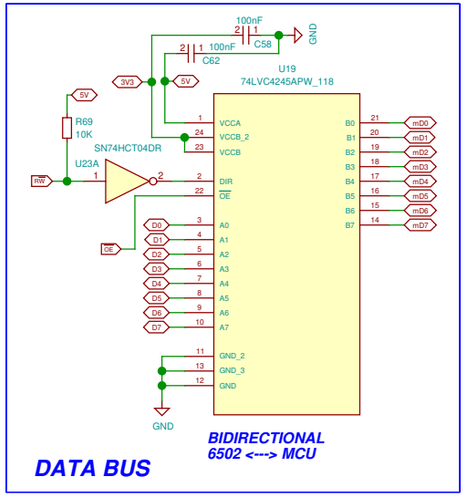




BUS LOGIC LEVEL SHIFTERS

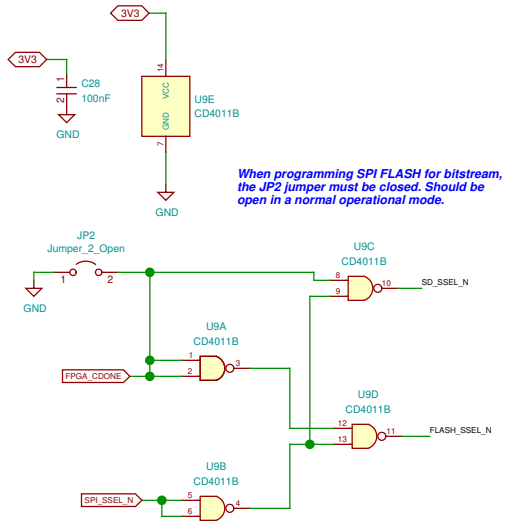




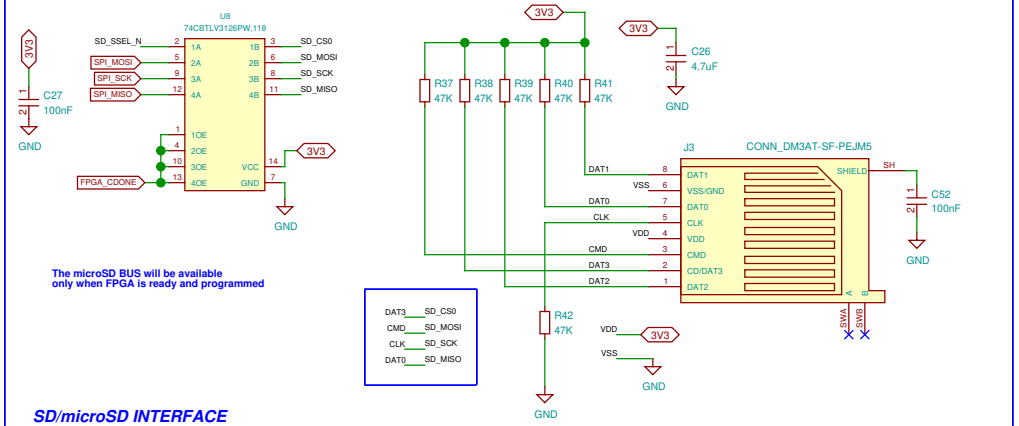


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Sheet: /CartridgeInterface/		
File: cartridgeInterface.sch		
Title: CartridgeInterface		
Size: A3	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261029-20:38Z		Id: 4/7

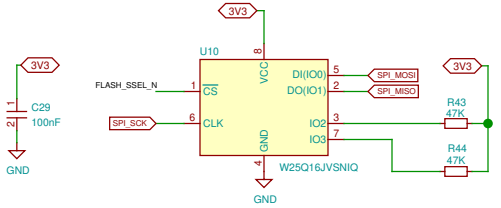
FPGA/SSD Flash Glue Logic



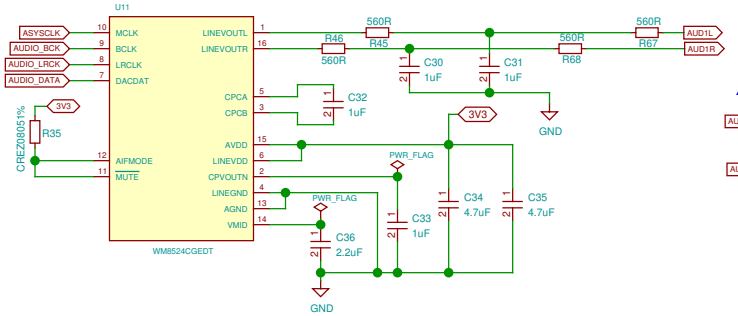
SD/microSD INTERFACE



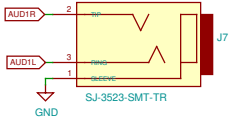
SPI 16MB FLASH



IC DAC/AUDIO 24BIT 192K 16TSSOP



AUDIO 3.5mm OUTPUT



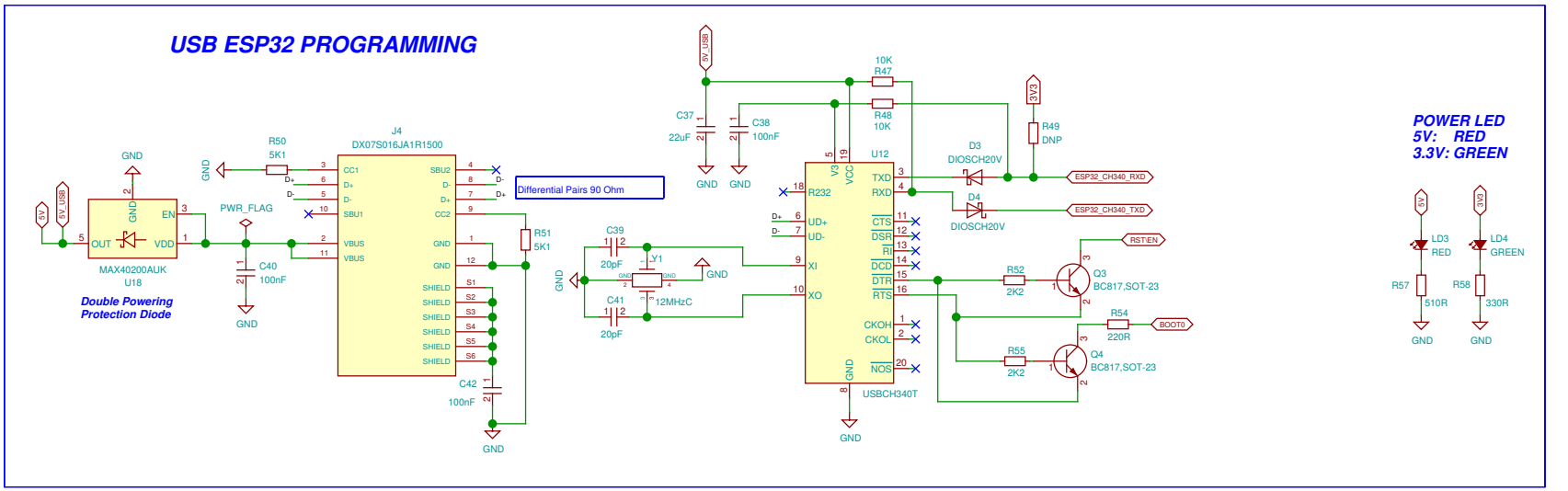
AUDIO SECTION

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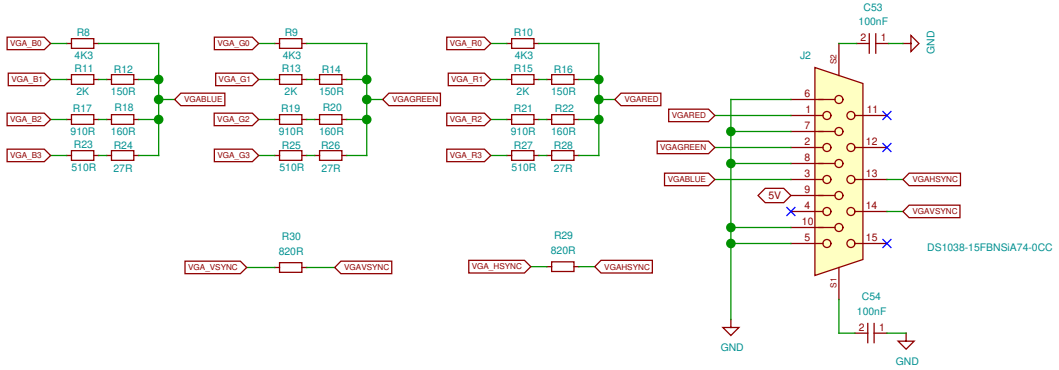
Sheet: /Vera FPGA flash/
File: vera-fpga-flash.sch

Title: Vera FPGA flash

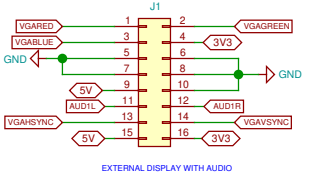
Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261029-20:38Z		Id: 5/7


USB ESP32 PROGRAMMING
5V_USB
10K
R47
R48
10K
3V3
R49
DNP
C37
22uF
C38
100nF
GND
GND
D3
DIOSCH20V
D4
DIOSCH20V
ESP32_CH340_RXD
ESP32_CH340_TXD
RST\EN
BOOT0
R52
2K2
Q3
BC817,SOT-23
R54
220R
R55
2K2
Q4
BC817,SOT-23
U12
R232
UD+
UD-
XI
XO
VCC
TXD
RXD
CTS
DSR
RI
DCD
DTR
RTS
CKOH
CKOL
NOS
USBCH340T
C39
20pF
Y1
12MHzC
C41
20pF
R50
5K1
J4
DX07S016JA1R1500
CC1
CC2
SBU1
SBU2
VBUS
SHIELD
R51
5K1
C42
100nF
Differential Pairs 90 Ohm
U18
MAX40200AUK
Double Powering
Protection Diode
PWR_FLAG
C40
100nF
POWER LED
5V: RED
3.3V: GREEN
LD3
RED
LD4
GREEN
R57
510R
R58
330R
POWER INPUT: from 5VDC...24VDC
Positive CENTRAL PIN
PJ-002AH-SMT-TR
J5
U15
L7805
C47
10uF-35V
C48
10uF-20V
R56
CREZ08051%
C49
100nF
U17
MAX40200AUK
Double Powering
Protection Diode
TP1
TP5V
POWER 3.3V & POWER 1.2V
U13
AMS1117-3.3
C44
100nF
C45
10uF-20V
C46
100nF
U14
AMS1117-1.2
C50
100nF
C51
10uF-20V
C43
1uF
1V2
Gianluca Renzi
RetroBit Lab
Sheet: /PowerSupply/
File: powersupply.sch
Title: PowerSupply
Size: A4
Date: 2026-10-26
Rev: 1.0
Plotted by eeshow 3a954ac+ 20261029-20:38Z
Id: 6/7

ANALOG VGA SIGNALS



EXTERNAL VIDEO CONNECTOR



Gianluca Renzi		
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Sheet: /VGA Analog/		
File: vga-analog.sch		
Title: VGA Analog		
Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261029-20:38Z		Id: 7/7